



SERIES — S

12V

SODIUM-ION BATTERY

Sodium-Ion NFPP technology engineered for RV, marine, off-grid, and mobility applications — with superior safety and maintenance-free operation.

**30–
100^{Ah}**

CAPACITY
RANGE

12^V

NOMINAL VOLTAGE

3,000+

CYCLE LIFE

40%

LIGHTER VS.
LEAD-ACID

–40/+60^{°C}

OPERATING TEMP

Zero

MAINTENANCE

coulomb
TECHNOLOGY

SODIUM ION BATTERY SERIES-S 12V

— ENERGY STORAGE YOU CAN BANK ON

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UN 38.3

IEC 62619

RoHS

CE

Air-Shippable

12V Sodium-Ion Battery Pack



12V 30 / 60 / 100Ah Sodium-Ion Battery Pack

The ultimate 12V battery solution for RV, marine, off-grid, and mobility applications. Available in 30Ah, 60Ah, and 100Ah capacities to match your power needs. Sodium-ion chemistry delivers superior safety, extreme temperature performance, and maintenance-free operation — making it the ideal drop-in replacement for lead-acid batteries.

COMPLETE BATTERY SYSTEM INCLUDES:

- Sodium-ion prismatic cells (4S configuration)
- Rugged weatherproof enclosure
- State-of-charge LED indicator
- Integrated BMS with protective circuitry
- Standard 12V terminals with protective covers
- Over-voltage & short-circuit protection

KEY BENEFITS



Safety-First Design

Sodium-ion chemistry eliminates fire risks associated with lithium-ion batteries — enabling safe use in enclosed spaces.

- Zero thermal runaway risk
- Non-hazardous air shipping
- No toxic gas emissions
- Safe for enclosed compartments



Superior Performance

Outperforms both lithium-ion and lead-acid in demanding conditions — from arctic cold to desert heat.

- 3,000+ cycles at 80% DoD
- -40°C to +60°C operation
- 40% lighter than lead-acid
- Fast charging capability



Maintenance-Free

True drop-in replacement requiring no special equipment or ongoing maintenance — just install and use.

- No watering or equalization
- No sulfation or memory effect
- Works with standard 12V chargers
- Built-in BMS protection

WHERE THE SERIES S PERFORMS



RV & MOTORHOME

Lights, fans, water pumps, and 12V appliances for extended off-grid camping trips.



MARINE & BOATING

Trolling motors, fish finders, and navigation equipment in harsh marine environments.



OFF-GRID & SOLAR

Cabins, sheds, and remote power setups with solar charge controller compatibility.



MOBILITY & SCOOTERS

Electric wheelchairs, scooters, and mobility devices with consistent power delivery.

3

CAPACITY OPTIONS (30/60/100AH)

12V

NOMINAL VOLTAGE

3,000+

CYCLE LIFE @ 80% DOD

-40°C

MINIMUM OPERATING TEMP

TECHNICAL SPECIFICATIONS — ALL MODELS

PARAMETER	30AH MODEL	60AH MODEL	100AH MODEL
Model Number	COL-12-30	COL-12-60	COL-12-100
Nominal Voltage	12V	12V	12V
Capacity	30 Ah	60 Ah	100 Ah
Energy	360 Wh	720 Wh	1,200 Wh
Cell Configuration	4S3P	4S6P	4S10P
Dimensions (L×W×H mm)	196×128×220	264×167×220	328×172×220
Weight	6.5 kg / 14.3 lb	9 kg / 19.8 lb	13 kg / 28.7 lb
Cycle Life	≥3,000 @ 1C 80% DoD / ≥5,000 @ 0.2C		
Charge Temperature	-10°C to +45°C		
Discharge Temperature	-40°C to +60°C		
Voltage Range	8V - 15.8V		
Max Discharge Rate	1.0C continuous		
Typical Runtime	12-24 hrs	1-2 days	2-4 days
Best For	Trolling motors, portable power	RV accessories, solar systems	House batteries, full RV setups



30Ah Model
COL-12-30 · 360 Wh



60Ah Model
COL-12-60 · 720 Wh



100Ah Model
COL-12-100 · 1,200 Wh

Sodium-Ion NFPP vs. Lead-Acid

A Clear Advantage Across Every Performance Dimension

Coulomb Technology's Sodium-Ion NFPP chemistry delivers superior cycle life, wider operating temperature range, zero maintenance, and a fundamentally safer chemistry — making it the ideal replacement for lead-acid in RV, marine, off-grid, and mobility applications.

FEATURE	SODIUM-ION NFPP	LEAD-ACID
Cycle Life	3,000+ cycles @ 80% DoD	300–500 cycles @ 50% DoD
Weight (100Ah)	13 kg (28.7 lb)	~32 kg (70.5 lb)
Usable Capacity	80–100% DoD	50% DoD (to preserve life)
Discharge Temp Range	–40°C to +60°C	–20°C to +50°C
Charge Temp Range	–10°C to +45°C	0°C to +40°C
Maintenance	Zero maintenance	Regular watering & equalization
Self-Discharge	<3% per month	5–15% per month
Charging Efficiency	~95%	~75%
Toxic Gas Emissions	None	Hydrogen gas during charging
Memory Effect	None	Sulfation if undercharged
Lifespan	10+ years	3–5 years
Environmental Impact	Non-toxic, recyclable	Lead & acid — hazardous

3 WAYS TO CHARGE

 12V Battery Charger Use any standard 12V sodium-ion or LiFePO4-compatible charger. Fully charges in 2–10 hours depending on capacity.	 Solar Panels (≥100W) Pairs seamlessly with any 12V solar charge controller. Recommended minimum 100W panel for efficient charging.	 Generator + DC Charger Connect a 12V DC charger to any AC generator for reliable charging when solar or shore power is unavailable.
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10× Longer cycle life vs. lead-acid	40% Lighter weight per 100Ah	Zero Maintenance required	95% Round-trip efficiency
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Built for Life Off the Grid

The Series S 12V battery delivers reliable, maintenance-free power wherever you need it — from weekend camping trips to full-time off-grid living. Its extreme temperature tolerance and superior cycle life make it the ideal drop-in replacement for lead-acid in any 12V application.



RV & MOTORHOME

RV & Camping Power

Power lights, fans, water pumps, and 12V appliances for extended off-grid camping trips. The lightweight design reduces vehicle weight compared to lead-acid batteries.

- Reliable power for extended off-grid trips
- Lightweight — reduces vehicle weight
- Works with existing 12V systems



MARINE & BOATING

Marine & Boat Power

Power trolling motors, fish finders, and navigation equipment in harsh marine environments. Vibration-resistant and corrosion-free with no toxic gas emissions.

- Safe for enclosed boat battery compartments
- Vibration-resistant construction
- No toxic gas emissions



OFF-GRID & SOLAR

Off-Grid Solar Storage

Small solar setups for cabins, sheds, and remote power. Deep discharge capability maximizes usable energy and pairs seamlessly with solar charge controllers.

- Deep discharge — 80–100% usable capacity
- Compatible with any solar charge controller
- 10+ year design life for remote installs



MOBILITY & ACCESSIBILITY

Mobility & Scooters

Power electric wheelchairs, scooters, and mobility devices with consistent power delivery throughout the full discharge cycle and improved range from reduced weight.

- Consistent power throughout discharge
- Lighter weight improves range
- Safe for indoor use — no gas emissions